Topological Embedding of the SHZ-BCC $\mathbf{Z}_2^3$ Symmetry in Loop Quantum Gravity

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Abstract

The Horizon Boundary Consistency (SHZ-BCC) mechanism relies on an 8-dimensional discrete channel space (Z_2^3) at the bifurcate horizon joint to constrain and cancel divergent vacuum energy. In this note, we formally embed this algebraic structure into the framework of Loop Quantum Gravity (LQG). We show that the $Z_2^{(s)} \times Z_2^{(n)} \times Z_2^{(p)}$ boundary channels map directly onto the intertwiner states of $SU(2)$ spin networks puncturing the causal horizon. This identification provides a rigorous UV-complete topological origin for the SHZ-BCC framework.

1 Introduction

In the effective SHZ-BCC model, the cosmological constant problem is addressed by demanding that physical observables on a 2D horizon joint $\Sigma = \mathcal{H}^+ \cap \mathcal{H}^-$ are invariant under arbitrary relabelings of binary horizon parameters: orientation (s), null direction (n), and polarization (p). The resulting finite group $G_{BCC} = Z_2^3$ enforces a projection that cancels the Λ_{bare} source. A critical open question is the microscopic origin of this discrete symmetry within a theory of quantum gravity.

2 LQG Spin Networks and the Isolated Horizon

In Loop Quantum Gravity, the classical spatial geometry is replaced by discrete spin networks, represented by graphs Γ with edges labeled by $SU(2)$ representations (spins j) and vertices labeled by intertwiners. When a spatial slice intersects a causal horizon Δ, the edges of the spin network puncture the horizon. The horizon itself is modeled as a $U(1)$ Chern-Simons surface theory.

Each puncture p carries a spin quantum number j_p, and the local geometry of the horizon is dictated by the eigenvalues of the area operator:

$$\hat{A}_\Sigma |\{j_p\}\rangle = 8\pi\gamma\ell_P^2 \sum_p \sqrt{j_p(j_p+1)}\, |\{j_p\}\rangle, \tag{1}$$

where γ is the Barbero-Immirzi parameter and ℓ_P is the Planck length.

3 Embedding the 8-Channel Space

To establish the correspondence between SHZ-BCC and LQG, we focus on the fundamental $j = 1/2$ spin network punctures. An elementary puncture in the vector representation provides a fundamental 2D Boolean degree of freedom (a "qubit" of area), often associated with the eigenvalues $m = \pm 1/2$ of the operator $\hat{J}_z$.

The SHZ-BCC joint Σ requires a multi-puncture correlation. We define the minimal non-trivial vertex structure crossing the bifurcate horizon as a **3-valent intertwiner node** interacting with the boundary.

3.1 The Isomorphism

Let a fundamental horizon boundary degree of freedom be described by the tensor product of three $j = 1/2$ spins localized at the joint:

$$\mathcal{H}_{joint}^{LQG} = \mathcal{V}_{1/2} \otimes \mathcal{V}_{1/2} \otimes \mathcal{V}_{1/2} \cong \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2. \tag{2}$$

This LQG Hilbert space has dimension $2^3 = 8$, matching exactly the channel space of the SHZ-BCC framework:

$$\dim \mathcal{H}_{ch}^{BCC} = \dim \left(\mathbb{C}_s^2 \otimes \mathbb{C}_n^2 \otimes \mathbb{C}_p^2\right) = 8. \tag{3}$$

We can rigorously map the SHZ-BCC parameters to the LQG observables:

- **Side/Orientation (s):** Maps to the sign of the flux operator $\hat{E} \cdot \hat{n}$ across the puncture, representing the inward/outward edge orientation.
- **Null Direction (n):** Maps to the choice of the gauge-fixing basis in the local $SU(2) \to U(1)$ reduction on the isolated horizon.
- **Polarization (p):** Maps to the intrinsic $\hat{J}_z$ eigenvalue (the $m = \pm 1/2$ states of the incident spin edge).

4 Vacuum Cancellation via SU(2) Gauge Invariance

In LQG, physical states must be gauge-invariant, requiring the sum of the fluxes at a vertex to vanish (the Gauss constraint): $\sum \hat{J}^{(i)} = 0$. For a state lying entirely on the horizon joint to be a physical observable, it must form a gauge-invariant singlet.

The SHZ-BCC group-averaging projector Π_{BCC} over the Z_2^3 channels acts exactly as a discrete topological projection onto the $SU(2)$ singlet state $|u\rangle$:

$$\Pi_{BCC}|\Psi\rangle = \frac{1}{8} \sum_{g \in Z_2^3} \hat{U}_g |\Psi\rangle = |u\rangle_{singlet}. \tag{4}$$

Because the bare vacuum energy acts as an active operator over the distinct non-gauge-invariant edge combinations, the enforcement of the LQG Gauss constraint (the singlet projection) mathematically replicates the SHZ-BCC cancellation mechanism $\Lambda_{bare}(1 - 2\chi) = 0$.

5 Conclusion

The ad-hoc 8-channel algebra of the Horizon Boundary Consistency model is not arbitrary. It represents the minimal discrete topology of a three-punctured $j = 1/2$ spin network crossing an isolated horizon in Loop Quantum Gravity. This profound connection bridges the phenomenological success of the SHZ-BCC flavor mechanism with a rigorous, UV-complete theory of quantum geometry.